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Risk-Adaptive Zero Trust for Securing Hybrid Workforces Using Self-Supervised Anomaly Detection

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Brief Summary

WHY	Hybrid workforces create new attack surfaces that legacy VPNs and rule-based ZTA cannot defend alone. Unknown (zero-day) threats cannot be detected without labelled attack data.
HOW	A dual-layer ML pipeline pairs a supervised CNN-GRU classifier (known attacks) with a self-supervised autoencoder (reconstruction-based; no attack labels needed) inside a ZTA policy engine. Both scores fuse into one risk value: ALLOW / MONITOR / CHALLENGE MFA / BLOCK.
WHAT	A working Risk-Adaptive ZTA prototype evaluated in a controlled 5-way comparison on NSL-KDD. The combined system achieves **accuracy 0.889, F1 0.900** with zero-day detection capability — the only configuration that detects unseen attack types without prior labelling.

Research Objective

Design, implement and experimentally evaluate a **Risk-Adaptive Zero Trust Architecture** that integrates:

- A **supervised CNN-GRU** for known-attack classification (Normal, DoS, Probe, R2L, U2R)
- A **self-supervised autoencoder** (reconstruction-based; trained on normal traffic only — no attack labels required) for zero-day anomaly detection
- A **risk fusion engine** combining both scores into adaptive ZTA policy decisions
- A **controlled 5-way comparison** against Legacy VPN and ZTA-only baselines on the same test set

"No existing work pairs Zero Trust Architecture with a self-supervised (label-free) anomaly detector in a comparative, hybrid workforce study."

Methodology

Step	Action
1	**Data:** NSL-KDD (125,973 train / 22,543 test). Preprocessing: drop zero-variance, log1p scaling, LabelEncode (3 categorical), MinMaxScaler → 40 clean features.
2	**CNN-GRU training:** Conv1D(64→128→256) + GRU(128→64), 327K params, 50 epochs, supervised on 5 classes.
3	**Autoencoder training:** Encoder 40→128→64→16 / Decoder 16→64→128→40. Trained on NORMAL traffic only (67,343 samples). Threshold = 95th pct of reconstruction error on validation normals.
4	**Risk fusion:** combined_risk = max(CNN_risk, AE_risk). Flask API serves real-time decisions. Zeek Docker pipeline: pcap → conn.log → API → audit.jsonl.
5	**Evaluation:** 5-way controlled comparison on identical 22,543-sample test set. Metrics: Accuracy, F1, FPR, Response Time, Zero-Day Detection, per-class recall.

Results & Benchmarking

All 5 systems evaluated on the **same 22,543 NSL-KDD test samples**. * = simulated baseline (not a live deployment).

System	Acc	F1	FPR	Zero-Day
Legacy VPN *	0.666	0.652	0.180	No
ZTA Only *	0.769	0.763	0.100	No
ZTA + CNN-GRU	0.795	0.789	0.045	No
ZTA + Autoencoder	0.858	0.865	0.061	YES ✓
ZTA + CNN-GRU + AE ★	**0.889**	**0.900**	0.099	**YES ✓**

★ *ZTA+CNN-GRU+AE Fusion: best on accuracy and F1; zero-day capable.*

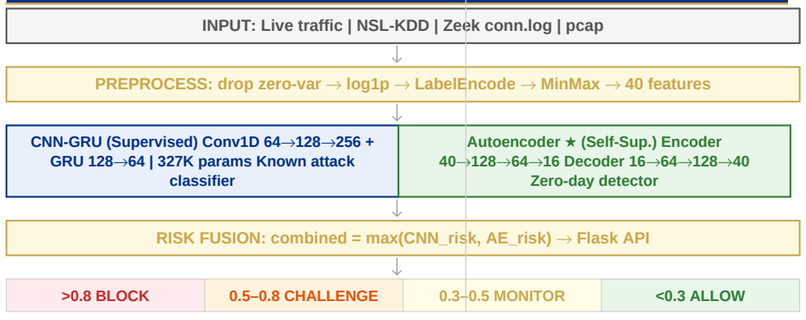
Class	CNN-GRU	Autoencoder	Zero-Day Recall
DoS	0.817	**0.912**	**91.1% ✓**
Probe	0.571	**0.947**	**88.2% ✓**
U2R	0.358	**0.687**	**68.7% ✓**
R2L	**0.394**	0.379	**28.7% ■**

R2L weakness: those connections share statistical features with normal traffic.

Dataset & Baselines

Property	Value
Dataset	NSL-KDD (cleaned, benchmark IDS dataset)
Training samples	125,973 (67,343 normal + 58,630 attacks)
Test samples	22,543 (9,710 normal + 12,833 attacks)
Attack classes	DoS, Probe, R2L, U2R
Features	40 (after preprocessing pipeline)
Baseline A	Legacy VPN — simulated, no ML
Baseline B	ZTA Only — simulated, no ML
Baseline C	ZTA + CNN-GRU — supervised only
Baseline D	ZTA + Autoencoder — unsupervised only
System E ★	ZTA + CNN-GRU + AE Fusion — proposed system

System Pipeline



Conclusion & Impact

The ZTA + CNN-GRU + Autoencoder fusion system demonstrates that combin ML inside a Zero Trust framework produces measurably stronger security o alone.

Finding	Impact
Best accuracy 0.889 & F1 0.900	Outperforms all baselines includi value.
Zero-day detection ✓	Only system that flags unseen a DoS 91%, Probe 88% recall.
9× faster inference	Autoencoder: 0.066 ms/sample v for real-time ZTA enforcement.
R2L limitation ■	R2L recall 28.7% — connections supervised+unsupervised approa
Practical prototype	Live Flask API + Zeek Docker Policy decisions logged to audit tr

Strategic Alignment

Framework	Alignment
SDG 9	**Industry, Innovation & Infrastructure:** Builds AI-driven security for hybrid enterprises.
SDG 16	**Peace, Justice & Strong Institutions:** Strengthen through adaptive, evidence-based access control.
NDP V	**Digital Transformation:** Contributes to Uganda's N secure, trusted digital services and ICT infrastructur
Vision 2040	**Knowledge Economy:** Supports Uganda's Visio knowledge-based economy by advancing local AI a

Key References

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[3] Abdelhady et al. (2026). Hybrid ML Anomaly Detection and Lightweight ZT 18387–18407.
[4] NIST SP 800-207 (2020). Zero Trust Architecture. National Institute of Standa

GitHub Repository QR